



Source-free domain adaptation for cross-domain remaining useful life prediction: A distributed federated learning perspective

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ABSTRACT

As the complexity of industrial equipment continues to increase, determining the remaining useful life (RUL) with high precision holds substantial significance for maintaining intricate industrial systems. The development of cross-domain prognostic approaches without source domain data necessitates thorough investigation, given the inherent distribution shifts among edge devices' degradation patterns and the imperative of preserving data security protocols. Furthermore, convolutional neural network, and long short-term memory network perform insufficiently when processing complex structurally dependent data. Consequently, this paper proposes a distributed RUL prediction approach based on graph convolutional neural network. Specifically, this paper designs a differential attention graph convolutional neural network that can focus on key areas in degradation data. Furthermore, considering the privacy and security of degradation data, this paper designs a two-stage decision boundary adjustment approach to achieve source-free RUL prediction under cross-domain conditions. On this basis, the study introduces a federated consensus mechanism that implements progressive weight calibration aligned with distributed training dynamics in edge computing environments, which can effectively reduce overfitting, and improve the generalization ability. Experimental validation on NASA's publicly available aircraft engine degradation dataset confirms the operational efficacy of the proposed approach.

1. Introduction

Complex systems such as modern energy equipment, aerospace, and intelligent manufacturing are characterized by high coupling, strong time-variability, and cross-scale collaboration [1–4]. Conventional point-based maintenance models are unable to support comprehensive safety and reliability maintenance [5–7]. With the deep integration of sensing, control, and information infrastructure, operations and maintenance are evolving from fault alerts to an integrated closed loop of prognosis and decision-making [8–10]. In this sense, the growing integration of automated components in industrial assets necessitates reliable remaining useful life (RUL) prediction, which directly enhances failure prevention capabilities while enabling cost-efficient maintenance planning for intricate production systems [11–13]. Consequently, the RUL has become indispensable for mission-critical industrial domains including aviation, power generation, and advanced production systems [14–17]. Conducting research on the RUL prediction not only has a profound impact on improving equipment management levels and maintenance efficiency, but also provides significant support for realizing industrial intelligence, and sustainable development [18–20].

Based on a large amount of historical data, the data-driven RUL prediction approaches adopt data mining technology to use rich data collected by monitoring equipment, to analyze the performance trends to predict the RUL of the system [21–23]. Notably, the approaches including convolutional neural networks (CNN) and long short-term memory networks (LSTM) boost RUL prediction performance by extracting data features. Zhu et al. proposed a contrastive bidirectional LSTM approach, which designs series parity decomposition to construct degradation data from multiple perspectives for the promotion of RUL prediction performance [24]. Ren et al. introduced a Transformer network with dynamic time length, capable of adaptively learning varying-length sequence representations for RUL prediction [25]. It is worth mentioning that CNN is expert at processing data with local spatial structure such as images. LSTM is expert at processing serialized data. Nevertheless, these approaches are insufficient in handling the dependencies of complex irregular data. As a deep learning architecture, Graph neural network (GNN) processes graph-based data and efficiently uncovers hidden non-Euclidean patterns. To enhance RUL prediction accuracy, Zhou et al. developed a multi-view spatio-temporal graph attention network capable of quantifying prediction uncertainty [26].

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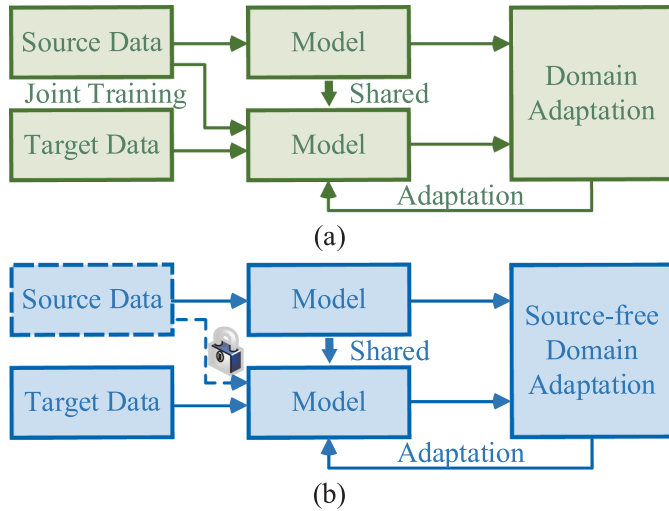


Fig. 1. (a) Conventional domain adaptation: adapts with source and target data. (b) Source-free domain adaptation: adapts with unlabeled target only (source data unavailable).

It is worth mentioning that in engineering practice, the industrial equipment is usually distributed in different geographical locations, which has notable limitations in data collection and centralized processing [27]. Furthermore, the device data may involve the core business, and confidential information of the factory or enterprise, so that the privacy must be strictly protected [28]. This integration of federated learning and distributed computing gives rise to distributed federated learning (DFL) as a timely solution. Each factory or enterprise is regarded as a client of the federation, so that the DFL performs local model training on each client. Moreover, the model parameters of each client are aggregated to avoid centralized data transmission, thus ensuring data privacy and security. In this sense, DFL enables efficient utilization of distributed device data, enhancing both model generalization and prediction accuracy [29]. Recent studies have demonstrated some approaches to RUL prediction using DFL frameworks. Guo et al. developed a federated learning framework incorporating convolutional autoencoders, which is trained collaboratively by cloud and edge devices to solve the problem of weak computing power [30]. Without the need for data sharing, Kamei et al. designed a Transformer and LSTM network architecture for DFL, which solves the challenges faced by traditional centralized learning [31]. Committed to solving the problems of data privacy and over-parameterization, taking bearings as the research object, Chen et al. established a DFL-based RUL prediction framework incorporating Taylor expansion pruning techniques [32].

There are significant domain differences in the operational environment, failure modes in different factories [33,34]. This domain drift phenomenon makes it difficult for the existing same-domain RUL prediction model to be directly applied to other domains [35,36]. Consequently, studying cross-domain RUL prediction techniques is imperative to achieve knowledge migration and exchange between domains, so as to be applicable to different types of industrial scenarios [37,38]. At present, some researchers have already carried out corresponding work. Chen et al. proposed an adaptive calibration approach based on transfer regression network, which can fully consider the shared and private characteristics of the target domain, thereby achieving accurate cross-domain RUL prediction [39]. It is worth mentioning that in a distributed federation, the factory or enterprise equipment represented by each client is affected by domain drift during operation [40].

Current DFL-based RUL prediction approaches seldom consider the domain drift phenomenon across different clients' data distributions. Moreover, most of the current conventional domain adaptation approaches demand joint training on source and target data (Fig. 1 (a)).

This joint training causes data privacy leakage. Consequently, how to achieve source-free domain adaptation (SFDA) (Fig. 1 (b)) requires in-depth work. The SFDA approach only needs a trained source model to improve target domain adaptation. The preceding discussion highlights the core research motivations for this work:

- Conventional CNN and LSTM are expert in processing local structured data. These approaches perform poorly when processing data with complex structural dependencies. Addressing these challenges through effective GNN implementation necessitates thorough methodological investigation [41–43].
- Industrial equipment is usually distributed in different geographical locations, which has notable challenges in protecting data privacy. How to study the RUL prediction problem under data privacy protection from the perspective of distributed federation is of great significance [40,44].
- Most existing RUL prediction approaches in DFL frameworks fail to account for cross-client domain shift in data characteristics. Furthermore, how to carry out SFDA-based RUL prediction more in-depth work [30–32].

To address these challenges, this work introduces a novel SFDA-assisted GNN approach within a DFL architecture for cross-domain RUL prediction. This research makes the following primary contributions:

- This paper proposes a differential attention graph convolutional network (DAGCN) architecture specifically designed to extract latent non-Euclidean features from equipment degradation patterns, which can focus on the key areas in the degradation data with the aid of attention mechanism. Furthermore, the graph information is aggregated during the training process to promote RUL prediction performance.
- To handle privacy-constrained domain shift in federated degradation data, this work proposes a dual-phase decision-boundary adaptation strategy that enables SFDA for cross-domain RUL prediction. By alternating discrepancy maximization and minimization from a source-pretrained initialization, the boundary is calibrated using only unlabeled target data without accessing any source samples.
- Extending SFDA principles, this work develops a distributed federated learning strategy with consistency-aware aggregation approach. Specifically, a dynamic weighting mechanism that down-weights drifting or unstable client is designed, which mitigates negative transfer and overfitting to individual clients, to improve robustness and accuracy for cross-domain RUL prediction in federated environments.

2. Problem formulation

This paper investigates source-free domain adaptation for cross-domain RUL prediction in DFL environments. The source domain data D_S can be expressed as $D_S = \{(x_i^S, y_i^S) | i = 1, 2, \dots, N_S\}$, x_i^S and y_i^S denotes the feature vector and corresponding RUL label of the i -th source domain sample, respectively, N_S is the number of source domain samples. The target domain data D_T can be expressed as $D_T = \{x_j^T | j = 1, 2, \dots, N_T\}$, where x_j^T denotes the feature vector of the j -th target domain sample, N_T denotes the target domain sample size. Considering the non-Euclidean features implicit in the degradation data of complex industrial systems, this paper adopts graph convolutional network (GCN) to mine features of degradation data. Specifically, the degradation data input to the GCN are expressed as $G = (X, V, A)$, where $X \in R^{n \times d}$ is the feature matrix of the graph nodes, d denotes the number of node channels, n denotes the number of nodes. In other words, X is the matrix after integrating the degradation data through the sliding time window. V denotes the vertex set of the graph, and $|V| = n$, $A \in R^{n \times n}$ denotes the adjacency matrix of the graph. If the two nodes v_i and v_j in the graph have a dependency relationship, it means that the

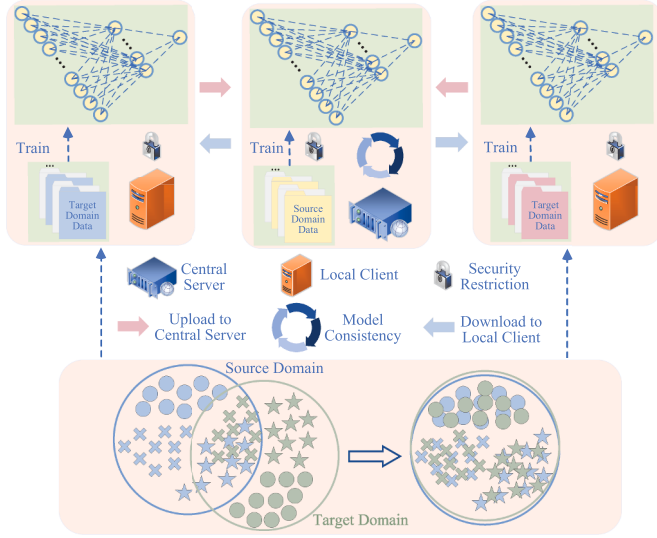


Fig. 2. The overall DFL-based framework for distributed cross-domain RUL prediction.

two nodes are connected to each other. At this time, $A_{ij} = 1$, otherwise $A_{ij} = 0$. This study tackles cross-domain RUL prediction using SFDA.

Furthermore, this paper considers the federated learning paradigm under data privacy protection requirements, a more general form of complex industrial systems. The degradation data of the source domain are only used to train a source model. In addition, each distributed federation is given this trained source model, which adopts unlabeled degradation data in the target domain for SFDA. In this sense, the distributed learning strategy is proposed to promote the network generalization ability. Consequently, the cross-domain RUL prediction is addressed through federated SFDA while guaranteeing client data confidentiality. The overall framework based on DFL is shown in Fig. 2.

3. SFDA For cross-domain RUL prediction based on DFL

3.1. Differential attention graph convolutional network

Graph convolutional network (GCN) is designed to handle graph-based data structures, which can well mine non-Euclidean features hidden in degradation data. Compared with conventional neural networks, CNN, LSTM, which are suitable for regular data, GCN can specifically handle irregular graph structure data. In detail, GCN learns node feature representations by propagating information on the graph structure. A node updates its representation by interacting with information from its neighbor nodes. This information sharing learning model is able to capture the local graph structure. In this sense, this paper designs an RUL prediction approach based on DAGCN. For the n -dimensional node feature matrix F^k input to the k th layer of the GCN, this paper designs a GCN based on the attention mechanism, shown as follows:

$$F^{k+1} = \text{Softmax} \left(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} F^k \Theta^k \right) \quad (1)$$

where I_n is the n -dimensional unit matrix, $\tilde{A} = A + I_n$, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$ is the diagonal matrix, Θ^k is the graph convolution layer parameter matrix, Softmax is the GCN activation function, which is used to calculate attention fraction. Committed to promoting the effective receptive field, and generalization ability of the network, GCN introduces pooling operations, which can focus on the most significant nodes or sub-graphs in the input graph, thereby decreasing computational complexity and overfitting.

This paper forms a node-feature difference matrix to measure how each node's features diverge from or align with its neighbors, highlighting crucial nodes in the graph. The feature difference matrix F_{diff} is

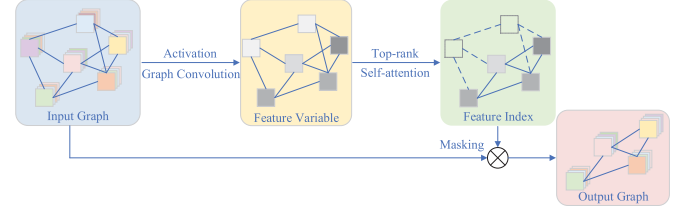


Fig. 3. The architecture of the proposed DAGCN for degradation representation learning.

shown in Eq. (2):

$$F_{diff} = \|F - \tilde{A}F\|_2^2 \quad (2)$$

On the basis of Eq. (2), the top-k sampling is expanded for the difference matrix to complete the differential attention operation, as shown in Eq. (3):

$$\begin{cases} idx = \text{top-rank}(F_{diff}) \\ \hat{F} = F_{idx} \\ \hat{A} = \tilde{A}_{idx,idx} \end{cases} \quad (3)$$

The function top-rank returns the top- k nodes by index, and idx specifies their index list. Applying the differential attention mechanism from Eqs. (2)–(3) yields the updated feature and adjacency matrices. After completing the differential attention mechanism, the convolution operation of the graph nodes is implemented similar to Eq. (1), and the above operation is repeated through the stacking of graph convolution layers. Lastly, the connection between graph convolution features and RUL is established with the aid of fully connected layers. After completing the model training process using source data, the source domain model is assigned to each federation client for SFDA. The structure of the DAGCN is shown in Fig. 3.

In the proposed approach, non-Euclidean degradation features are extracted in DAGCN by treating multi-sensor measurements as node features on a graph and performing topology-normalized message passing for neighborhood aggregation. Furthermore, this design improves interpretability by providing explicit node-level importance signals. The discrepancy scores quantify how strongly each sensor node deviates from its neighborhood context, and the selected index set identifies which sensors/substructures are emphasized during pooling.

Many GCN-based prediction models mainly rely on a fixed adjacency-driven neighborhood aggregation, with homogeneous message passing weights to capture local structural dependencies. The proposed DAGCN performs an attention-formulated convolution in which the activation is used to compute attention scores over graph-propagated features in Eq. (1). Moreover, it constructs a node-feature difference matrix to quantify how each node's features diverge from its neighbors in Eq. (2). Based on this discrepancy measurement, the model conducts top-rank sampling to retain the most informative nodes/sub-graphs, which masks the remaining ones to form an updated feature and adjacency representation in Eq. (3). This architecture-level difference-driven attention is fundamentally different from conventional GCN pipelines used for RUL prediction, because it does not merely aggregate neighborhood information but adaptively concentrates the receptive field on salient degradation-sensitive sensors or sub-structures. Differential attention with graph convolutions is topology-aware and discrepancy-driven, which is instead of weighting sensors by feature similarity as in self-attention, it highlights nodes whose representations deviate from their neighborhood after graph propagation. With graph convolution modeling local sensor dependencies, yielding more robust representations under domain shifts.

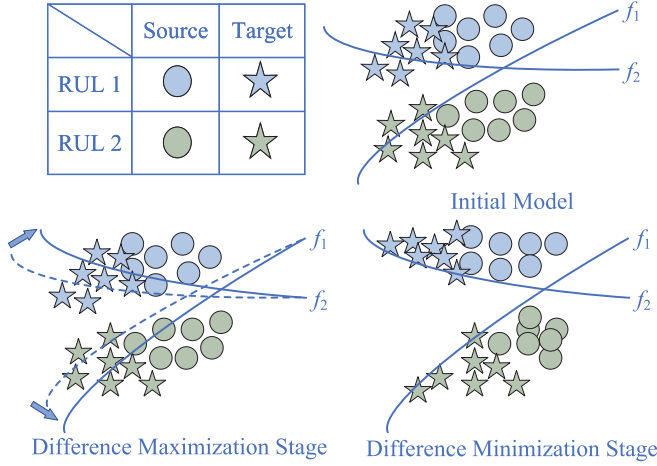


Fig. 4. The basic idea of two-stage decision boundary adjustment.

3.2. The proposed SFDA approach

From the structural level of the model, the DAGCN is composed of a differential attention graph convolution feature extraction sub-network (DAGCFN) and an RUL predictor. Specifically, the DAGCFN is composed of Eqs. (1)–(3). The RUL predictor consists of a fully connected layer network, mapping high-dimensional graph convolution features to RUL. Regarding source domain degradation data, this paper jointly trains the DAGCFN and the RUL predictor. The loss function for the source domain appears below:

$$L_S = \frac{1}{N_S} \sum_{i=1}^{N_S} (RUL_{pi} - RUL_{ti})^2 \quad (4)$$

where N_S is the number of source domain degradation samples, RUL_{pi} is the predicted RUL of the i -th source sample, and RUL_{ti} is the true RUL value of the i -th source sample.

After completing the model training process using source data, each client in the distributed federation can be assigned the source domain model for SFDA training. For each federated client, a DAGCFN, and two RUL predictor sub-networks are constructed, respectively. On this basis, this paper proposes a two-stage decision boundary adjustment approach, the basic idea of which is shown in Fig. 4. The comprehensive procedure is outlined as follows:

1) **Difference maximization stage:** According to the DAGCN model obtained from the source domain, each federated client can acquire a DAGCFN, and two RUL predictor sub-networks. The DAGCFN is fixed. Moreover, the RUL prediction difference is maximized between two RUL predictor sub-networks to detect source-supported target samples, which are far away from the decision boundary. Consequently, the loss function may be further formulated as shown in Eq. (5):

$$L_m^{(c)} = \frac{1}{N_T^{(c)}} \sum_{j=1}^{N_T^{(c)}} \left| f_1^{(c)} \left[g_{DAGCFN}^{(c)} \left(x_j^T \right) \right] - f_2^{(c)} \left[g_{DAGCFN}^{(c)} \left(x_j^T \right) \right] \right| \quad (5)$$

where $N_T^{(c)}$ denotes the quantity of target domain degradation samples for the c -th federated client, x_j^T is the j -th target sample, $f_1^{(c)}$ and $f_2^{(c)}$ are the mapping relationship of the first RUL predictor, and the second RUL predictor of the c -th federated client, respectively, $g_{DAGCFN}^{(c)}$ is the relationship of DAGCFN of the c -th federated client. In addition to constructing the RUL prediction value difference loss between the two RUL predictor sub-networks, this paper also designs a regularization strategy for the weights to prevent the predictors from diverging during the training process, shown as follows:

$$L_{reg,1}^{(c)} = \left\| \theta_S - \theta_{f_1^{(c)}} \right\|^2 + \left\| \theta_S - \theta_{f_2^{(c)}} \right\|^2 \quad (6)$$

On this basis, the first stage's loss function can be computed as given in Eq. (7):

$$\min_{f_1^{(c)}, f_2^{(c)}} L_{\max}^{(c)} = \left| \lambda_1 L_{reg,1}^{(c)} - L_m^{(c)} \right| \quad (7)$$

where λ_1 serves as the weighting factor between difference loss and regularization loss.

2) **Difference minimization stage:** After completing the first training stage, the two RUL predictor sub-networks are fixed. Furthermore, the DAGCFN's optimization process seeks to minimize the output variance between the dual RUL predictor sub-networks, so that the extracted features are more closed to the decision boundary. During this process, a weight regularization strategy must be implemented to maintain DAGCFN stability during training, shown as follows:

$$L_{reg,2}^{(c)} = \left\| \theta_{DAGCFN}^S - \theta_{DAGCFN}^{T(c)} \right\|^2 \quad (8)$$

where $\theta_{DAGCFN}^{T(c)}$ is the parameters of DAGCFN corresponding to the c -th federated client's target domain, and θ_{DAGCFN}^S is the parameters of DAGCFN corresponding to source domain. On this basis, the loss function of the second stage can be calculated as follows:

$$\min_{DAGCFN} L_{\min}^{(c)} = \lambda_2 L_{reg,2}^{(c)} + L_m^{(c)} \quad (9)$$

The training process of the whole two-stage decision boundary adjustment approach is shown in Fig. 5.

3.3. The DAGCN-SFDA-DFL framework for cross-domain RUL prediction

Fig. 6 depicts the proposed DAGCN-SFDA-DFL framework for cross-domain RUL prediction. Specifically, in the central server, a DAGCN model is constructed through Eqs. (1)–(3), where the loss function facilitates learning the mapping between source degradation data and RUL. Furthermore, the source domain model is assigned to each client in distributed federation for SFDA training.

Regarding each client, a DAGCFN and two RUL predictor sub-networks are constructed, respectively. According to the source DAGCN-based prediction model obtained from the central server, the model is split and assigned to DAGCFN, and two RUL predictor sub-networks in each client. On this basis, this paper proposes a two-stage decision boundary adjustment approach. 1) Difference maximization stage: The DAGCFN is fixed. Moreover, the RUL prediction difference is maximized between two RUL predictor sub-networks to detect source-supported target samples, which are far away from the decision boundary, shown in Eqs. (5)–(7). 2) Difference minimization stage: The two RUL predictor sub-networks are fixed. Additionally, the DAGCFN optimizes to reduce discrepancies between the dual RUL predictor sub-networks, so that the extracted features are more closed to the decision boundary, shown in Eqs. (8)–(9). By performing the above steps alternately, a single training process of each federated client is completed.

After completing the calculation of the training process in each federated client, it is necessary to acquire the corresponding evaluation indicators to guide the federated communication. Consequently, this paper designs an evaluation function in Eq. (10), whose essence is to evaluate the difference between each distributed federated model and the source domain model, shown as follows:

$$L_e^{(c)} = L_{\min}^{(c)} + L_{\max}^{(c)} \quad (10)$$

Additionally, the inter-client communication is established through a proposed prediction model consistency strategy, where each client uploads its trained DAGCFN and dual RUL predictor sub-networks to the central server. The central server performs parameter optimization using client model evaluation metrics, assigning each model a weight coefficient as formulated in Eq. (11):

$$w_c = \frac{1 - \frac{L_e^{(c)}}{\sum_{c=1}^C L_e^{(c)}}}{C - 1} \quad (11)$$

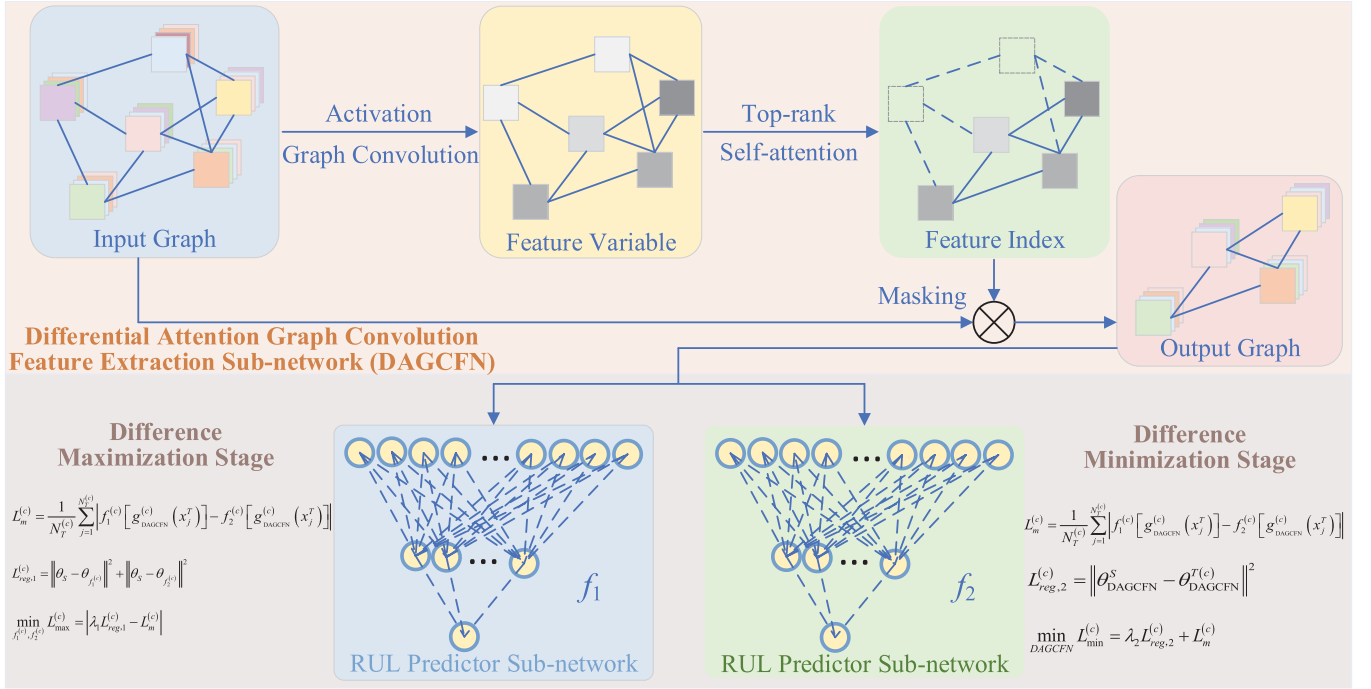


Fig. 5. The training process of the whole two-stage decision boundary adjustment approach.

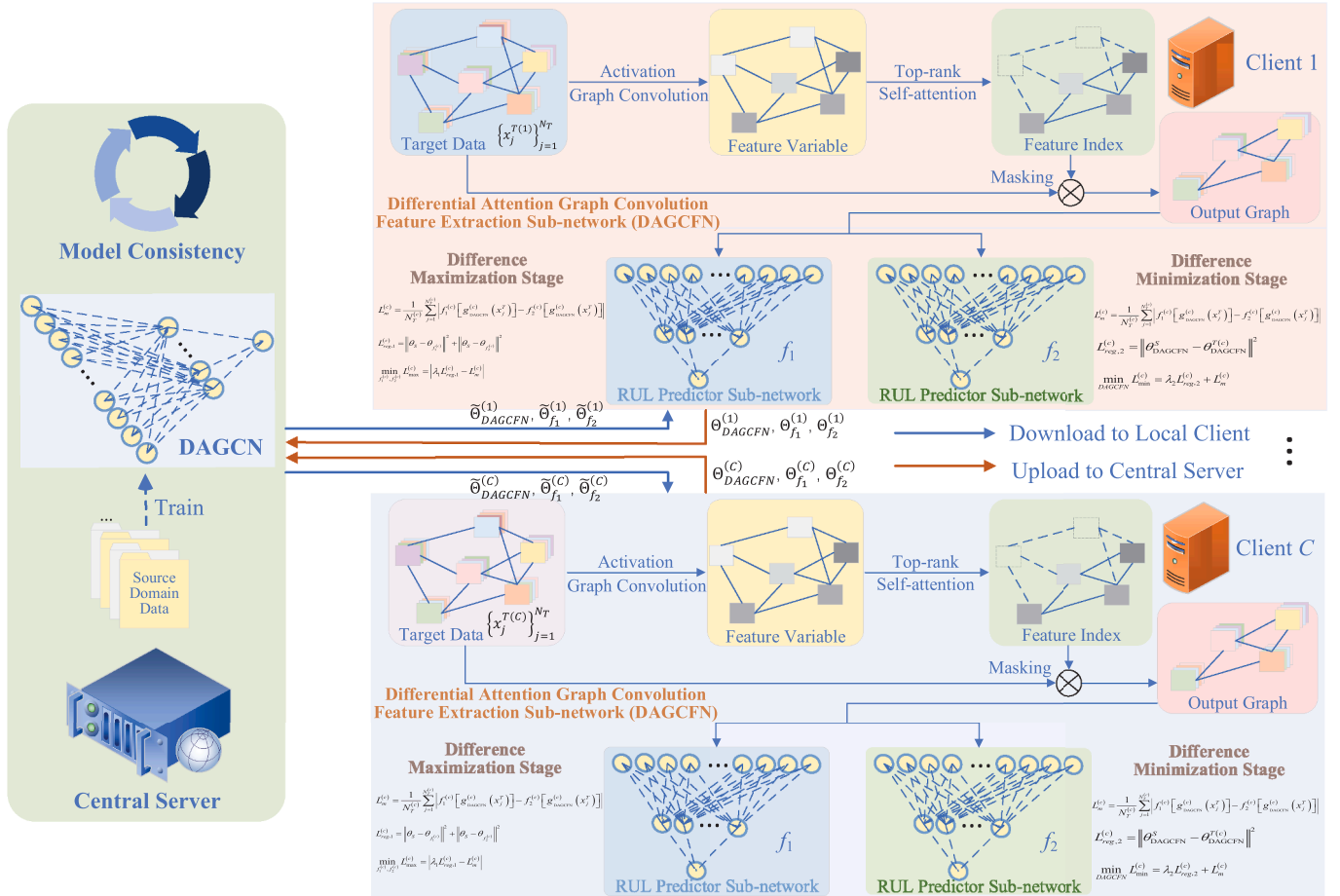


Fig. 6. The DAGCN-SFDA-DFL framework for source-free federated cross-domain RUL prediction, integrating DAGCN-based degradation representation, dual-phase boundary adaptation, and consistency-aware federated aggregation.

where C is the total client count, and w_c is the model weight coefficient the c -th client. According to client weighting factors, the model parameters of the DAGCFN $\tilde{\Theta}_{DAGCFN}$, and the two RUL predictor sub-networks $\tilde{\Theta}_{f_1}$, $\tilde{\Theta}_{f_2}$, can be obtained separately, shown as follows:

$$\tilde{\Theta}_{DAGCFN} = \sum_{c=1}^C \left(w_c \Theta_{DAGCFN}^{(c)} \right) \quad (12)$$

$$\tilde{\Theta}_{f_1}, \tilde{\Theta}_{f_2} = \sum_{c=1}^C \left(w_c \Theta_{f_1}^{(c)} \right), \sum_{c=1}^C \left(w_c \Theta_{f_2}^{(c)} \right) \quad (13)$$

The proposed model consistency strategy accounts for inter-client parameter variations by incorporating federated network parameters, thereby enhancing generalization while mitigating overfitting risks. Moreover, the approach prevents cross-client leakage of target domain data, rigorously maintaining data privacy. After obtaining the DAGCFN and two RUL predictor sub-networks' model parameters, the central server disseminates the enhanced models to all clients for additional training epochs. This iterative process continues until prediction errors converge across all federation clients. Online model evaluation is efficiently executable upon training convergence in this architecture. For individual clients, online target domain data are segmented into time windows and simultaneously input to the trained DAGCFN and its two RUL prediction branches. The online RUL can be obtained as follows:

$$RUL_{pk}^{(c)} = \frac{1}{2} \left\{ f_1^{(c)} \left[g_{DAGCFN}^{(c)} \left(x_k^{T(c)} \right) \right] + f_2^{(c)} \left[g_{DAGCFN}^{(c)} \left(x_k^{T(c)} \right) \right] \right\} \quad (14)$$

where $RUL_{pk}^{(c)}$ is the RUL prediction value of the k -th online target sample of the c -th client, and $x_k^{T(c)}$ is the k -th online target sample of the c -th client.

It should be noted that the proposed federated consensus mechanism follows a synchronous, round-based client-server protocol rather than asynchronous communication. In each round, the server broadcasts the current global model to all participating clients, waits for their local updates, and then performs weighted aggregation only after receiving the uploads from the round's clients, before proceeding to the next round. Moreover, convergence is ensured through a federation-level stopping criterion and stability-aware optimization. Specifically, training proceeds for a fixed maximum number of communication rounds or terminates early when the validation/prediction error stabilizes across rounds. In addition, the alternating two-stage boundary adjustment with source regularization mitigates client drift under non-IID shifts, reducing oscillations and promoting stable convergence.

Unlike FedAvg, which performs uniform parameter averaging and may suffer from negative transfer when clients undergo heterogeneous source-free adaptation, the proposed aggregation is guided by a model-consistency criterion coupled to the two-stage boundary adjustment. Specifically, after each client completes the difference maximization/minimization procedure, Eq. (10) is designed as an evaluation signal reflecting the stability of boundary adjustment and the degree of deviation from the source prior. The server converts $L_e^{(c)}$ into normalized weights w_c , so that clients with more consistent adaptation contribute more to the global model, while clients with larger drift are down-weighted, which helps mitigate inter-client parameter divergence and improves cross-client generalization under privacy-preserving cross-domain RUL prediction.

This approach is suitable for edge deployment. Specifically, the differential attention performs top- k selection, which reduces the effective graph size for subsequent layers and thus further lowers computation. In practice, inference is run on low-power devices, while federated adaptation can run on edge gateways or during idle periods. Communication only transmits model updates, and computational complexity can be reduced by decreasing the number of rounds.

3.4. The network optimization for SFDA based on DFL

The proposed two-stage decision boundary adjustment approach can be divided into a difference maximization stage and a difference minimization stage. In these two phases, it is necessary to perform weight updates separately for the network parameters in each federated client. Specifically, for the difference maximization stage, the network model optimization problem can be shown as follows:

$$\hat{\Theta}_{f_1}^{(c)}, \hat{\Theta}_{f_2}^{(c)} = \arg \min_{\Theta_{f_1}^{(c)}, \Theta_{f_2}^{(c)}} E \left[\Theta_{DAGCFN}^{(c)}, \Theta_{f_1}^{(c)}, \Theta_{f_2}^{(c)} \right] \quad (15)$$

where $\hat{\Theta}_{f_1}^{(c)}$, $\hat{\Theta}_{f_2}^{(c)}$ are the optimal parameter values of the two RUL predictors of the c -th federated client. At each training iteration, the dual RUL predictors undergo parameter updates as follows:

$$\Theta_{f_1}^{(c)} = \Theta_{f_1}^{(c)} - \eta \frac{\partial L \left[\Theta_{DAGCFN}^{(c)}, \Theta_{f_1}^{(c)}, \Theta_{f_2}^{(c)} \right]}{\partial \Theta_{f_1}^{(c)}} \quad (16)$$

$$\Theta_{f_2}^{(c)} = \Theta_{f_2}^{(c)} - \eta \frac{\partial L \left[\Theta_{DAGCFN}^{(c)}, \Theta_{f_1}^{(c)}, \Theta_{f_2}^{(c)} \right]}{\partial \Theta_{f_2}^{(c)}} \quad (17)$$

where η is the learning rate. On the other hand, for the difference minimization stage, the network model optimization problem can be expressed as follows:

$$\hat{\Theta}_{DAGCFN}^{(c)} = \arg \min_{\Theta_{DAGCFN}^{(c)}} E \left[\Theta_{DAGCFN}^{(c)}, \Theta_{f_1}^{(c)}, \Theta_{f_2}^{(c)} \right] \quad (18)$$

where $\hat{\Theta}_{DAGCFN}^{(c)}$ is the optimal value of the DAGCFN parameters for the c -th federated client. At each training iteration, the DAGCFN undergoes parameter updates as follows:

$$\Theta_{DAGCFN}^{(c)} = \Theta_{DAGCFN}^{(c)} - \eta \frac{\partial L \left[\Theta_{DAGCFN}^{(c)}, \Theta_{f_1}^{(c)}, \Theta_{f_2}^{(c)} \right]}{\partial \Theta_{DAGCFN}^{(c)}} \quad (19)$$

By minimizing the loss function of the two stages defined, the DAGCFN, and two RUL predictors of each distributed federated client are optimized. The domain invariance in degradation data can be learned in an adversarial manner. The DFL-based aggregation strategy enables more comprehensive utilization of federated client resources. This privacy-preserving approach simultaneously strengthens client models' generalization capacity, and reduces overfitting potential, leading to superior RUL prediction performance. In the implementation, privacy is ensured in the standard federated learning sense that raw degradation data never leave local clients, and only model parameters/updates are communicated to the server. Furthermore, the main objective of this paper is to improve cross-domain RUL prediction performance through source-free adaptation and distributed federated training. Therefore, the modeling effectiveness is emphasized under domain shift and different clients rather than designing additional cryptographic privacy mechanisms.

It should be noted that the federated consensus is implemented in a synchronous, round-based protocol, which does not include a dedicated fault-tolerant communication scheme. However, it supports partial participation. In detail, if some clients are delayed or drop out, the server aggregates only the received updates and re-normalizes the consistency-aware weights over participating clients, which avoids stale and missing updates, and reduces sensitivity to client availability.

4. Case study

4.1. Dataset introduction and task description

The proposed DAGCN-SFDA-DFL is evaluated on NASA Ames' complete life-cycle dataset for aircraft turbofan engines. This dataset collects aircraft turbofan engine operational status and sensor data from four different operational mode (OM), which covers a wealth of feature

Table 1

The configuration of NASA aircraft engine dataset.

Dataset	FD001	FD002	FD003	FD004
Training engines	100	260	100	249
Testing engines	100	259	100	248
Operation modes	1	6	1	6
Failure modes	1	1	2	2

Table 2

Cross-domain RUL prediction tasks across heterogeneous OMs.

Task	Source domain	Target domain		
		Client 1	Client 2	Client 3
1	OM1	OM2	OM3	OM4
2	OM2	OM1	OM3	OM4
3	OM3	OM1	OM2	OM4
4	OM4	OM1	OM2	OM3

information. In this sense, the basic conditions are created to simulate the inconsistent distribution of each degradation dataset in a distributed federation. Table 1 presents the complete dataset specifications. This dataset contains multi-source sensor data of aircraft turbofan engines, such as temperature, pressure, a total of 26 variables. Each operational condition contains a certain scale of degradation data for the construction and verification of the RUL prediction model. The detailed description of the dataset can be found in [45].

Table 1 reveals notable data distribution discrepancies across the four scenarios, attributable to differing operational parameters and failure mechanisms in the aircraft engines. The proposed system is grounded in federated learning principles, enabling collaborative model training across decentralized clients. The central server trains a source domain model for a labeled source domain dataset, which assigns the trained model to each client. Each client has an unlabeled target domain degradation dataset. Consequently, this work addresses unsupervised RUL prediction by leveraging SFDA within a privacy-preserving federated framework. To validate the approach's generalizability, the RUL prediction is performed across heterogeneous OMs and cross-domain configurations, with comprehensive results tabulated in Table 2. Experiments are conducted on the NASA turbofan engine dataset, using all four subsets FD001-FD004. For cross-domain validation, each subset is treated as an independent domain. In each task, one subset serves as the labeled source domain for initial training, while the remaining subsets are treated as unlabeled target domains for source-free adaptation, and final performance is reported on the official test split of each target subset using the provided ground-truth RUL.

4.2. Implementation details of the experiment

Some feature variables are constant during the operation of an aircraft turbofan engine, which are not useful in RUL prediction and need to be removed. Therefore, seven types of constant sensor data numbered 1, 5, 6, 10, 16, 18, and 19 are excluded. Normalization of the processed features improves both the convergence rate and training robustness of the model. This paper adopts the maximum-minimum normalization method for processing. Consistent with the formulation in Section II, time-windowing is applied to sensor data to preserve temporal industrial system dynamics for deep learning.

It should be noted that the learning rate η , the time window length, the number of layers of DAGCN, the number of neurons in each layer of the DAGCN, the number of fully connected layers, batch size, and weight decay, may have an impact on the RUL prediction performance of the proposed approach. This section changes the value of learning rate η , the time window length, the number of layers of DAGCN, the number of neurons in each layer of the DAGCN, the number of fully connected layers, batch size, and weight decay to numerically study

Table 3

The grid search parameter optimization details.

Parameter	Range	Configuration
Learning rate	[0.001, 0.005, 0.01, 0.02]	0.01
Time window length	[10, 20, 30, 40, 50, 60]	30
DAGCN layer	[1, 2, 3, 4, 5]	3
DAGCN neuron	[8, 16, 32, 64, 128]	16
Fully connected layer	[1, 2, 3, 4, 5]	2
Batch size	[32, 64, 128, 256, 512]	256
Weight decay	$[10^{-8}, 10^{-7}, 10^{-6}, 10^{-5}, 10^{-4}]$	10^{-7}

Table 4

Hyperparameter configuration of the whole network.

Parameter	Configuration
Learning rate	0.01
Time window length	30
Epoch	50
Number of DAGCN layer	3
Number of DAGCN neuron	16
Fully connected layer	2
Batch size	256
Weight decay	10^{-7}
Loss function	Eq. (10)
Optimizer	Adam

the robustness of these hyperparameters. In this study, OM1 is treated as the source domain, while OM2-OM4 are assigned to three different clients as target domains. To strictly comply with the source-free setting, all hyperparameters are selected only on a held-out validation split of the OM1 training set (without using any labeled target data) and then fixed across all cross-domain transfers and federated clients to avoid any target-specific tuning. This paper adopts a grid parameter adjustment strategy by traversing parameter settings. As can be seen from Table 3, when the above parameters are 0.01, 30, 3, 16, 2, 256, and 10^{-7} , respectively, the RUL prediction performance in the source domain is the best. On this basis, the neural network parameter configuration is shown in Table 4. To enhance reproducibility and practical implementation, a sensitivity analysis is further conducted for all hyperparameters listed in Table 3 by varying one factor within its reasonable range while keeping the others unchanged. The results indicate that the proposed approach remains stable over a broad interval and shows noticeable degradation mainly at extreme values, supporting the robustness of the framework to hyperparameter choices.

4.3. The design of ablation experiment

To evaluate the marginal contribution of each core component, this paper constructs three ablation variants, all of which are run under the same data partitioning, preprocessing, training rounds, and random seeds as the main experiment, with only the ablated modules being modified.

M1 (Replacing DAGCN with GCN): The DAGCN's topology-aware propagation and differential aggregation are replaced with standard GCN, maintaining the same number of layers and hidden dimensions. The remaining modules (SFDA and distributed training strategy) remain unchanged.

M2 (Removing two-stage domain adaptation training): All target domain self-training and divergence, and boundary adjustment mechanisms of source-free domain adaptation are disabled on the DAGCN, which adopts only source domain pre-trained weights for direct inference in the target domain. The remaining modules remain unchanged.

M3 (Removing distributed federated learning): In the proposed configuration, the cross-client consistent aggregation and weight fusion are disabled, which performs only single-client (or single-scenario) local training. The remaining modules remain unchanged.

Table 5
The RMSE on the aircraft engine dataset under ablation variants.

Task	M1	M2	M3	Ours
OM1-OM2	22.8	30.7	30.1	20.6
OM1-OM3	26.4	34.9	33.6	24.5
OM1-OM4	28.7	36.8	34.9	26.6
OM2-OM1	16.0	24.8	22.3	14.2
OM2-OM3	27.2	34.9	31.6	26.1
OM2-OM4	33.4	37.6	32.9	31.8
OM3-OM1	18.6	29.2	24.2	17.1
OM3-OM2	37.1	43.5	36.5	30.6
OM3-OM4	34.2	39.8	38.9	31.0
OM4-OM1	20.1	28.8	27.7	18.2
OM4-OM2	21.9	29.9	27.9	20.4
OM4-OM3	19.3	32.0	29.5	17.5
Mean	25.5	33.6	30.8	23.2

The RMSE on the aircraft engine dataset under ablation variants are shown in Table 5. The ablations reveal a stable importance ranking of components: two-stage source-free domain adaptation > distributed federated learning > DAGCN's structural inductive bias. Removing SFDA (M2) produces the largest degradation across tasks, which underscores that, under unlabeled target data, divergence-driven self-training and boundary adjustment are the primary forces that absorb inter-domain shift, and prevent negative transfer. Disabling the distributed strategy (M3) yields the next-largest drop, indicating that cross-client consistency, and weight aggregation are crucial to control variance and avoid overfitting to idiosyncratic sub-domains. The effect is most visible when operating-mode heterogeneity is pronounced. Replacing DAGCN with a plain GCN (M1) leads to the smallest losses. Notably, per-task fluctuations show occasional M1-M3 cross-overs, which is expected given differing degrees of covariate shift versus structural mismatch, but the mean ranking remains unchanged, and the full model dominates on every task.

4.4. The approach comparison experiment and discussion

As detailed in Table 2, this section implements the DFL-based cross-domain RUL prediction framework across all specified operational scenarios. Comparative experiments highlight the superior performance of the proposed approach relative to existing methods. Specifically, this paper implements a DAGCN without DFL to compare the results, where the network model structure is the same. Performance advantages are shown by benchmarking against existing aircraft engine cross-domain approaches. Table 6 presents the performance comparison across approaches. The table also indicates whether the approach is an SFDA approach. Furthermore, OM1 is designated as the source domain, whereas OM2-OM4 are allocated to three clients as targets. Figs. 7–9 represent the RUL prediction results of DAGCN-SFDA-DFL, and DAGCN without DFL. Table 6 and Figs. 7–9 demonstrate the superior performance of the DAGCN-SFDA-DFL RUL prediction approach.

To make the stage-wise behavior more intuitive and directly observable, qualitative trajectory-level evidence is represented for the OM1-OM2 transfer task by plotting the RUL predicted value and the ground-truth over the full life cycles of four representative OM2 test engines (IDs 14, 50, 138, and 148), which is shown in Fig. 10. The visualization shows that the proposed approach generally maintains great consistency between predicted and actual RUL throughout the whole lifecycle, while minor deviations may occur in the mid-life stage due to the instability and time-varying nature of the degradation process, thereby directly addressing lifecycle consistency under a concrete OM1-OM2 cross-domain task.

Experimentally, the proposed approach delivers consistently improved cross-domain RUL prediction, and the distributed setting further amplifies this gain (Tables 5 and 6), indicating stronger generalization under domain shift. Methodologically, the two-stage SFDA boundary adjustment refines the target decision boundary using unlabeled tar-

get data while avoiding overconfident, target-specific fitting, and the consistency-aware federated aggregation down-weights unstable client updates, reducing client-specific overfitting and improving robustness under non-IID heterogeneity.

The comparative results in Tables 5 and 6 and the prediction results in Figs. 7–10 jointly evidence three complementary advantages of the proposed DAGCN-SFDA-DFL framework: accuracy under shift, stability across heterogeneous clients, and credible generalization without target labels. The advantages can be summarized as follows:

(1) Compared with recently published cross-domain turbofan approaches and with the internal variants (M1-M3), the proposed approach achieves uniformly lower errors, which indicates that the decision boundary learned on the source domain has been effectively adapted to target distributions.

(2) The two-stage source-free domain adaptation is the primary driver of robustness. By combining divergence-driven self-training with boundary adjustment, the model absorbs both covariate and conditional shift without any target labels. This mitigates negative transfer, particularly on OM3 and OM4 where operational mode heterogeneity and dual failure patterns are more pronounced, so that the target predictions remain close to the true degradation. The qualitative improvement is visible in Figs. 7–9, where DAGCN-SFDA-DFL tracks the ground truth more tightly than the non-DFL counterpart.

(3) The distributed federated learning contributes consistent gains in the federated setting by suppressing client drift. Cross-client consistent aggregation aligns local optima and improves the worst-case client. This translates into a practical advantage: better generalization under non-IID target partitions while preserving data locality and privacy (no raw signal sharing). Moreover, non-IID heterogeneity is handled by using consistency-aware weighted aggregation. In this sense, stable clients are up-weighted and drifting clients are down-weighted, which suppresses negative transfer and prevents performance degradation.

(4) The graph-structured inductive bias of DAGCN complements SFDA and DFL. By modeling directed sensor interdependencies and emphasizing failure-critical subgraphs, DAGCN's structural prior sharpens the model's focus on informative channels and preserves performance when the target domain exhibits topology-dependent changes.

(5) The approach scales to hundreds of clients since learning is local. Moreover, the server only performs weighted aggregation (linear in participating clients), with cost further reduced by partial participation. The communication depends on model size, not raw data. Furthermore, the sensor heterogeneity is handled by graph-based modeling with masking or sub-graph operation, while top- k pooling improves robustness to varying sensor sets.

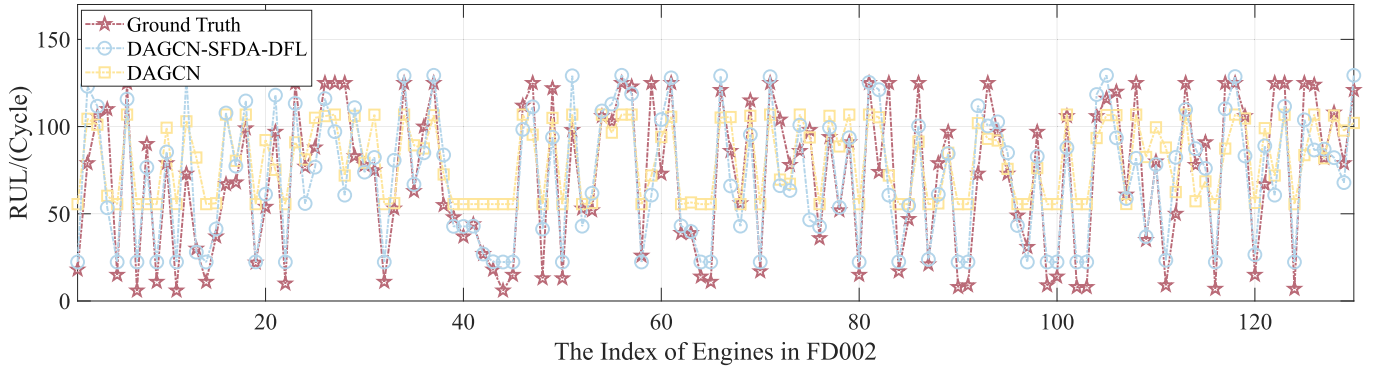
Taken together, these components explain why the full DAGCN-SFDA-DFL pipeline consistently outperforms both single-domain adaptation approaches and federated baselines that lack SFDA. The proposed approach adapts where it matters (decision boundaries), aggregates what is reliable (client-consistent updates), and represents what the system truly is (interacting sub-systems). Moreover, the framework is tailored to multi-sensor time-series, where sensors naturally define interpretable graph nodes. For images and text logs, the key limitation is the lack of an explicit topology and clear node semantics, while SFDA and federated aggregation remain unchanged.

It should be noted that the proposed approach is compatible with real-time edge inference, because online deployment only requires a forward pass of a lightweight DAGCN on a sparse sensor graph, with computation scaling mainly with the number of edges, and selected nodes. In addition, the discrepancy-guided top- k pooling further reducing the effective graph size. In practice, inference can run on low-power devices, while SFDA and federated adaptation can be executed on edge gateways or periodically when devices are idle. In other words, for resource-constrained edge devices, continuous inference can run locally, while federated adaptation can be executed on edge gateways or scheduled during idle periods to balance latency, energy, and bandwidth budgets.

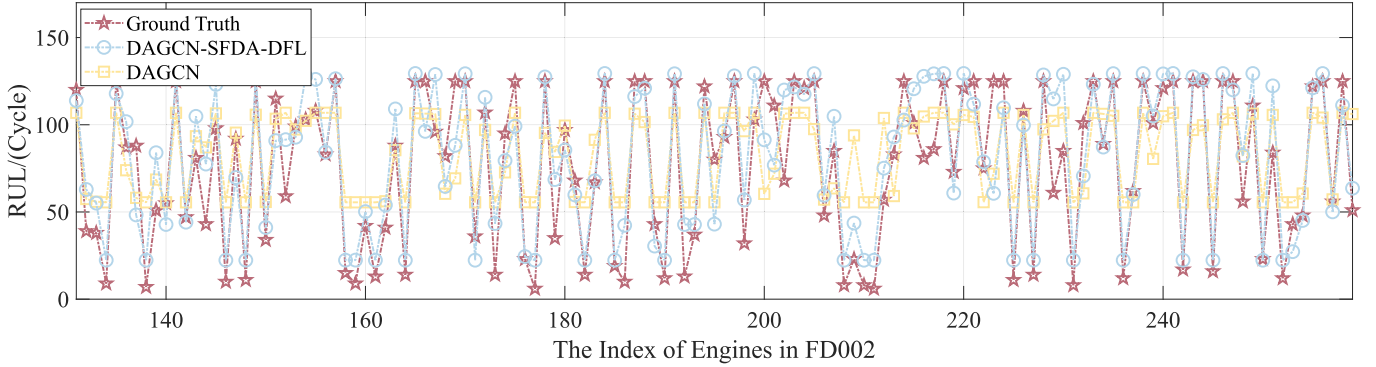
Table 6

The compared experimental RMSE for aircraft engine dataset.

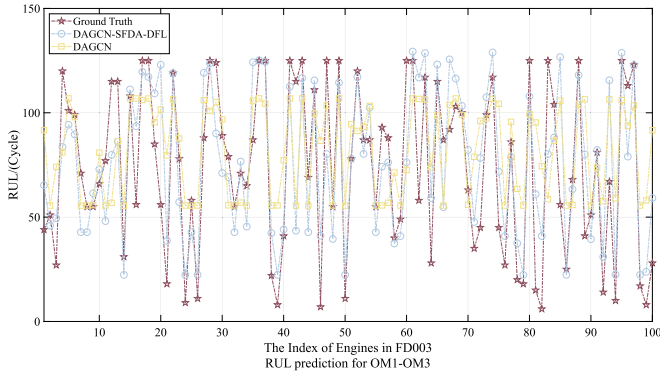
Task	DNN-TCA [46]	DNN-CORAL [46]	DANN-CNN [46]	SFMCD[35]	DA [47]	NoDF [47]	MRAN- LSTM [48]	DANN- LSTM [46]	CURA[49]	M3	Ours
Source-free	×	×	×	✓	×	×	×	×	✓	✓	✓
OM1-OM2	90.0	77.5	50.3	29.0	49.4	44.7	–	48.6	32.2	30.1	20.6
OM1-OM3	116.1	69.6	42.8	42.2	28.3	24.3	–	45.9	39.0	33.6	24.5
OM1-OM4	113.8	84.6	87.0	61.0	52.5	46.4	–	43.8	37.4	34.9	26.6
OM2-OM1	85.6	80.9	47.4	51.4	42.4	34.9	–	28.1	43.3	22.3	14.2
OM2-OM3	111.5	79.8	40.8	50.9	43.6	45.3	–	37.5	38.7	31.6	26.1
OM2-OM4	94.4	43.6	38.0	56.2	37.3	35.1	–	31.9	42.0	32.9	31.8
OM3-OM1	90.5	26.5	33.8	42.8	22.1	18.2	29.8	31.7	26.7	24.2	17.1
OM3-OM2	80.8	75.6	68.2	41.0	62.1	51.1	42.2	44.6	22.6	36.5	30.6
OM3-OM4	102.6	77.2	73.7	44.9	64.0	54.2	43.5	47.9	30.5	38.9	31.0
OM4-OM1	85.6	94.0	67.4	56.4	37.2	36.0	32.9	31.5	33.9	27.7	18.2
OM4-OM2	80.8	30.9	39.1	39.5	25.5	24.4	27.3	24.9	35.3	27.9	20.4
OM4-OM3	102.9	68.6	45.5	39.4	46.4	42.5	36.1	27.8	35.4	29.5	17.5
Mean	96.2	67.4	52.8	46.2	42.6	38.1	–	37.0	34.7	30.8	23.2



(a) RUL prediction for OM1-OM2 (Index: 1-130)



(b) RUL prediction for OM1-OM2 (Index: 131-259)

Fig. 7. The RUL prediction results for OM1-OM2. (x-axis: The index of engines, y-axis: RUL/(Cycle)).**Fig. 8.** The RUL prediction results for OM1-OM3. (x-axis: The index of engines, y-axis: RUL/(Cycle)).

The proposed approach is effective, when a meaningful sensor graph is available, the domain shift is moderate for label-free boundary refinement, and client participation is stable for aggregation. The performance may degrade under extreme mismatch, rapid drift, sensor dropout/reconfiguration (topology changes), highly heterogeneous clients (uncertainty/pseudo-label instability), or federated issues such as partial participation, stragglers, and low-quality/adversarial updates.

4.5. The visual analysis of features in neural network

This work examines DAGCN-SFDA-DFL's cross-domain feature characteristics through visualizations of DAGCN, and DAGCN-SFDA-DFL feature extractions. In detail, this paper takes the OM1-OM2 task as an instance. The probability density distributions of extracted features for DAGCN and DAGCN-SFDA-DFL are contrasted in Fig. 11. Divergent failure modes and operating conditions cause data distribution variations between OM1 and OM2. Fig. 11(a) reveals partial overlap yet

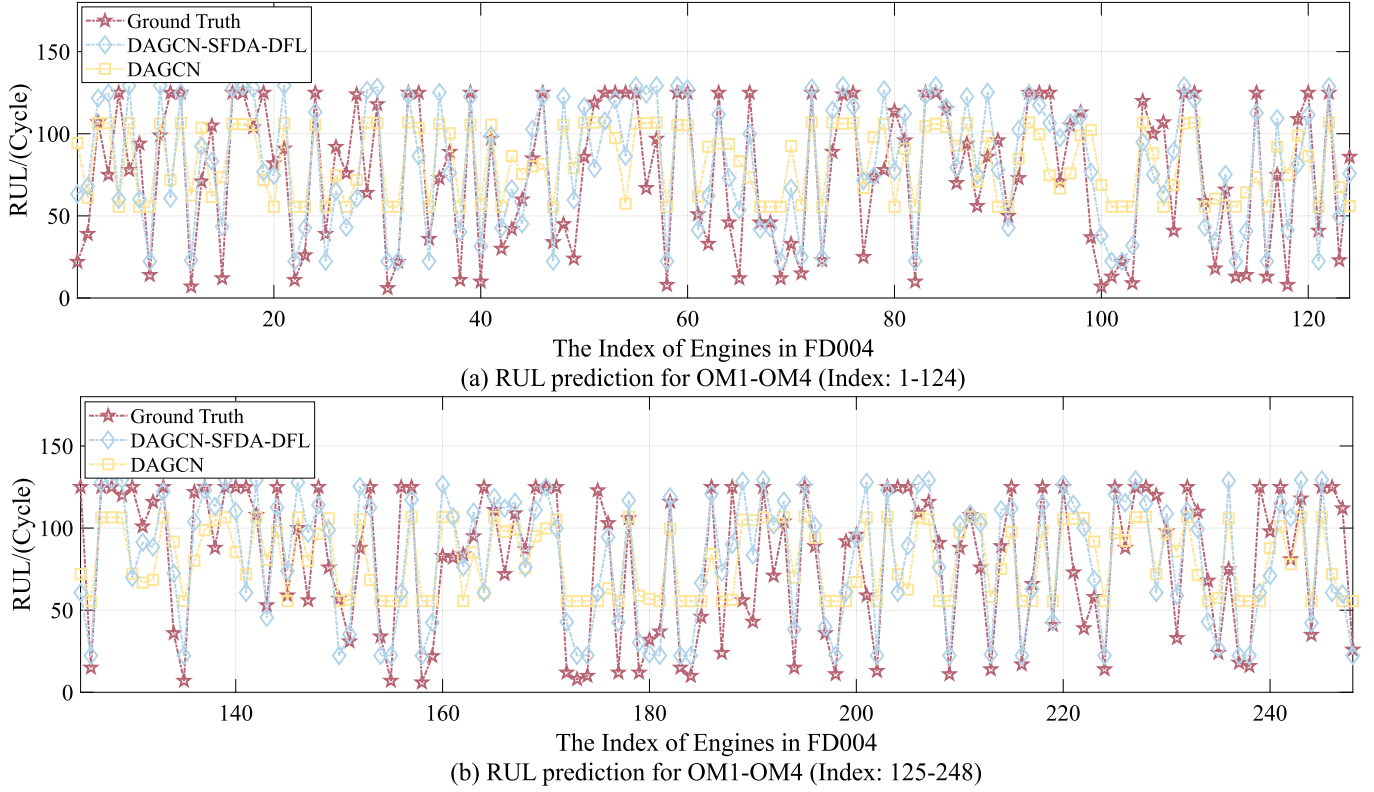


Fig. 9. The RUL prediction results for OM1-OM4. (x-axis: The index of engines, y-axis: RUL/(Cycle)).

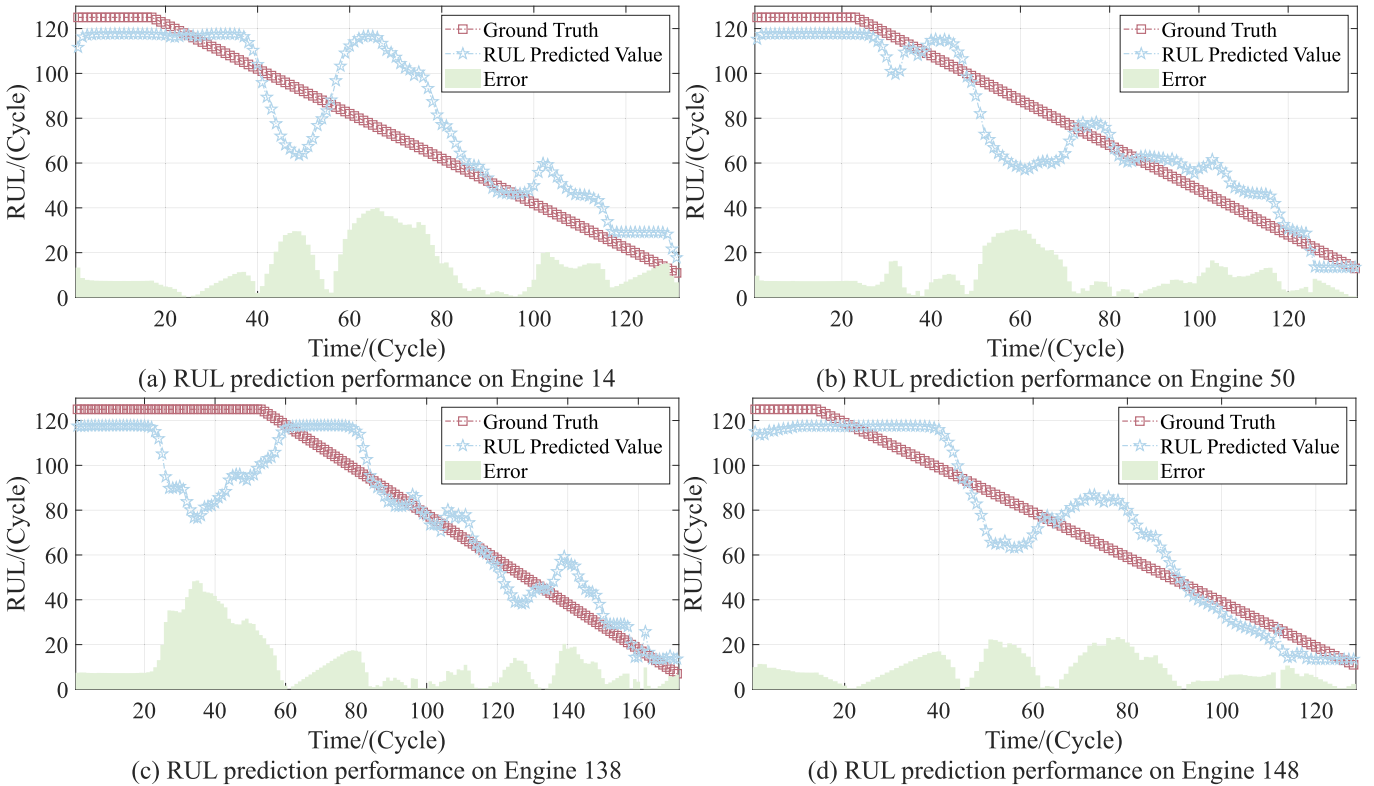


Fig. 10. Trajectory-level RUL prediction results on representative OM2 test engines under OM1-OM2. (x-axis: Time/(Cycle), y-axis: RUL/(Cycle)).

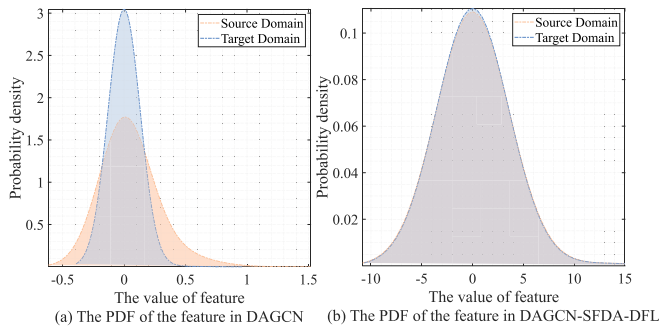


Fig. 11. The PDF of the source domain and target domain. (x-axis: The value of feature, y-axis: Probability density).

notable inconsistencies in high-dimensional feature distributions across operational conditions. These findings demonstrate the complexities in domain-adaptive RUL prediction. Direct application of source-domain information yields suboptimal target-domain performance. On the other hand, of particular significance, the DAGCN-SFDA-DFL proposed in this paper adopts an adversarial SFDA strategy with DFL, which has a great domain adaptation capability to align the source features and the target features. In this sense, an accurate mapping relationship from degradation features to RUL can be better established, which also confirms the comparison results in [Tables 5 and 6](#).

5. Conclusion

To extract non-Euclidean degradation patterns while preserving data privacy, a DAGCN-SFDA-DFL is proposed for cross-domain RUL prediction. Specifically, a DAGCFN and two RUL predictor sub-networks are designed for each client in the distributed federation. A model consistency strategy enhances generalization during interactive training. This research makes the following primary contributions: (1) This paper proposes a DAGCN architecture specifically designed to extract latent non-Euclidean features from equipment degradation patterns, which can focus on the key areas in the degradation data with the aid of attention mechanism. Furthermore, the graph information is aggregated during the training process to promote RUL prediction performance. (2) To overcome domain shift in distributed federated degradation data without compromising privacy, this work proposes a dual-phase decision boundary adaptation strategy enabling SFDA for cross-domain RUL prediction. (3) Extending SFDA principles, this work develops a distributed federated learning strategy that maintains model consistency through dynamic weight adjustment, reducing overfitting risks and improving cross-domain RUL prediction accuracy. The results show that the proposed approach achieves a mean RMSE of 23.2 over 12 transfer tasks, which outperforms strong baselines and internal variants. In particular, compared with the internal counterpart without the distributed federated mechanism, the mean RMSE is reduced from 30.8 to 23.2. Moreover, the ablation study removes two-stage domain adaptation training increases the mean RMSE to 33.6, which indicates that source-free decision-boundary refinement and reliability-weighted aggregation jointly drive the performance improvements. Some source-target pairs remain challenging under severe shifts, yielding higher errors due to weak alignment of target dynamics with the source-induced features/boundary and label-free adaptation issues. The proposed differential attention and consistency-aware aggregation mitigate but cannot fully eliminate such extreme mismatch effects. Future work will explore adaptive graph learning with time-varying sensor connectivity, and reinforcement-based boundary tuning, to further improve robustness under non-stationary degradation and heterogeneous federated clients. Moreover, future work will also broaden applicability by extending the framework to multi-modal sensing based on modality-

specific encoders, and multi-view graphs, and enabling real-time adaptation through label-free incremental boundary update with stability control.

CRedit authorship contribution statement

Jiushi Zhang: Writing – original draft, Validation, Methodology, Funding acquisition, Conceptualization; **Chunxiao Wang:** Validation, Software, Formal analysis; **Quan Qian:** Writing – review & editing, Funding acquisition; **Shen Yin:** Writing – review & editing.

Data availability

The data that has been used is confidential.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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